Fixing Missing Element Code in Python

https://gemini.google.com/app/5460edfba4cca14d

User prompt: I ran into some errors when solving a python problem. The goal was to find the missing number from an array of numbers. For example given array A such that: A[0] = 2 A[1] = 3 A[2] = 1 A[3] = 5the function should return 4, as it is the missing element. Here is my code: def solution(A): present_nums = set(A) n = len(A) for nums in range(1, n+1): if nums not in present_nums: return nums But it failed these correctness tests:

Response: The function returns `None` (causing the `<class 'NoneType'>` error) because **the range does not include the last possible element**.

In this problem (Codility's *PermMissingElem*), an array of length N contains distinct numbers from the range $[1 \dots N+1]$. Because exactly one number is missing, the highest possible value is $N+1$.

In Python, `range(1, n + 1)` stops before $n+1$, meaning it only checks values from 1 to n:

- If `A = []` ($N = 0$), the missing element is `1`. `range(1, 1)` does not run, and the function returns `None`.
- If `A = [1]` ($N = 1$), the missing element is `2`. `range(1, 2)` only checks `1`, finishes the loop, and returns `None`.
- Whenever the missing value is $N+1$, your loop finishes without hitting `return`, returning `None`.

Fix 1: Adjust the range bound

Expand the range to `n + 2` so it includes $N+1$:

```
def solution(A):
    present_nums = set(A)
    n = len(A)

    for num in range(1, n + 2):
        if num not in present_nums:
            return num
```

Fix 2: Arithmetic sum ($O(N)$ time, $O(1)$ space)

The sum of integers from 1 to $N+1$ is given by the formula $\frac{(N+1)(N+2)}{2}$. Subtracting the sum of array `A` directly yields the missing number without allocating a hash set:

```
def solution(A):
    n = len(A)
```

```
    expected_sum = (n + 1) * (n + 2) // 2
    return expected_sum - sum(A)
```
